

|   |                          |                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                               |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|----------------------|----------------------------------------|---|---|-----|-----|---|------|------|------|---|---|------|------|---|-------|---|-------|---|
| 4 | (a)                      | The <u>INCREASE</u> in the internal energy of a system is equal to the sum of heat supplied <u>TO</u> the system and the work done <u>ON</u> the system. |                                                                                                                                                                                                                                                                                                                                                                               | 1          |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
|   | (b)                      | (i)                                                                                                                                                      | <p>Pressure <math>p / 10^5 \text{ Pa}</math></p> <p>16</p> <p>7.8</p> <p>1.0</p> <p>1.4 6.0 10</p> <p>Volume <math>V / 10^{-4} \text{ m}^3</math></p>                                                                                                                                                                                                                         | 4          |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
|   |                          | (ii)                                                                                                                                                     | <table><tr><td></td><td>Heat supplied to gas / J</td><td>Work done on gas / J</td><td>Increase in internal energy of gas / J</td></tr><tr><td>A</td><td>0</td><td>300</td><td>300</td></tr><tr><td>B</td><td>2580</td><td>-740</td><td>1840</td></tr><tr><td>C</td><td>0</td><td>-440</td><td>-440</td></tr><tr><td>D</td><td>-1700</td><td>0</td><td>-1700</td></tr></table> |            | Heat supplied to gas / J | Work done on gas / J | Increase in internal energy of gas / J | A | 0 | 300 | 300 | B | 2580 | -740 | 1840 | C | 0 | -440 | -440 | D | -1700 | 0 | -1700 | 3 |
|   | Heat supplied to gas / J | Work done on gas / J                                                                                                                                     | Increase in internal energy of gas / J                                                                                                                                                                                                                                                                                                                                        |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
| A | 0                        | 300                                                                                                                                                      | 300                                                                                                                                                                                                                                                                                                                                                                           |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
| B | 2580                     | -740                                                                                                                                                     | 1840                                                                                                                                                                                                                                                                                                                                                                          |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
| C | 0                        | -440                                                                                                                                                     | -440                                                                                                                                                                                                                                                                                                                                                                          |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
| D | -1700                    | 0                                                                                                                                                        | -1700                                                                                                                                                                                                                                                                                                                                                                         |            |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |
|   |                          | (iii)                                                                                                                                                    | <p>Net work done on gas = <math>300 + (-740) + (-440) + 0 = -880 \text{ J}</math></p> <p>Heat supplied to gas during B = <math>2580 \text{ J}</math></p> <p>Ratio = <math>880 / 2580 = 0.34</math></p>                                                                                                                                                                        | 1<br><br>1 |                          |                      |                                        |   |   |     |     |   |      |      |      |   |   |      |      |   |       |   |       |   |

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|---|-----|--|--|
| 5 | (2) |  |  |
|---|-----|--|--|